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Systems and methods for processing objects including a linear gantry system

Abstract

A storage, retrieval and processing system for processing objects is disclosed. The storage, retrieval and processing system includes a plurality of storage bins providing storage of a plurality of objects, where the plurality of storage bins being in communication with a retrieval conveyance system, a programmable motion device in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins, where the programmable motion device includes an end effector for grasping and moving a selected object out of a selected storage bin, and is adapted for movement of the programmable motion device along a first direction, and a plurality of destination bins that are provided in at least one linear arrangement along the first direction of movement of the programmable motion device.

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Background/Summary

PRIORITY (1) The present application is a continuation of U.S. patent application Ser. No. 17/730,036, filed Apr. 26, 2022, which is a continuation of U.S. patent application Ser. No. 16/776,870, filed Jan. 30, 2020, now U.S. Pat. No. 11,358,794, issued Jun. 14, 2022, which is a continuation of U.S. patent application Ser. No. 15/923,765 filed Mar. 16, 2018, now U.S. Pat. No. 10,596,696, issued Mar. 24, 2020, which claims priority to U.S. Provisional Patent Application Ser. No. 62/473,082, filed Mar. 17, 2017, the disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

- (1) The invention generally relates to storage and retrieval systems, and relates in particular to automated storage and retrieval systems that are used with systems for processing objects.
- (2) Automated storage and retrieval systems (AS/RS) generally include computer controlled systems of automatically storing (placing) and retrieving items from defined storage locations. Traditional AS/RS typically employ totes (or bins), which are the smallest unit of load for the system. In these systems, the totes are brought to people who pick individual items out of the totes. When a person has picked the required number of items out of the tote, the tote is then re-inducted back into the AS/RS.
- (3) In these traditional systems, the totes are brought to a person, and the person may either remove an item from the tote or add an item to the tote. The tote is then returned to the storage location. Such systems, for example, may be used in libraries and warehouse storage facilities. The AS/RS involves no processing of the items in the tote, as a person processes the objects when the tote is brought to the person. This separation of jobs allows any automated transport system to do what it is good at—moving totes—and the person to do what the person is better at—picking items out of cluttered totes. It also means the person may stand in one place while the transport system brings the person totes, which increases the rate at which the person can pick goods.
- (4) There are limits however, on such conventional systems in terms of the time and resources required to move totes toward and then away from each person, as well as how quickly a person can process totes in this fashion in applications where each person may be required to process a large number of totes. There remains a need therefore, for an AS/RS that stores and retrieves objects more efficiently and cost effectively, yet also assists in the processing of a wide variety of objects.

SUMMARY

- (5) In accordance with an embodiment, the invention provides a storage, retrieval and processing system for processing objects. The storage, retrieval and processing system includes a plurality of storage bins providing storage of a plurality of objects, where the plurality of storage bins being in communication with a retrieval conveyance system, a programmable motion device in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins, where the programmable motion device includes an end effector for grasping and moving a selected object out of a selected storage bin, and is adapted for movement of the programmable motion device along a first direction, and a plurality of destination bins that are provided in at least one linear arrangement along the first direction of movement of the programmable motion device.
- (6) In accordance with another embodiment, the storage, retrieval and processing system includes a plurality of storage bins providing storage of a plurality of objects, where the plurality of storage bins being in communication with a retrieval conveyance system that includes automated means for providing selected storage bins to an input conveyance system, a programmable motion device in communication with the input conveyance system for receiving the selected storage bins from the plurality of bins, where the programmable motion device includes an end effector for grasping and moving a selected object out of each selected storage bin, and is adapted for movement of the programmable motion device along a first direction, and a plurality of destination bins being provided in at least one linear arrangement along the first direction of movement of the programmable motion device.
- (7) In accordance with a further embodiment, the invention provides a method of providing storage, retrieval and processing of objects. The method includes the steps of providing a plurality of storage bins for storing a plurality of objects, where the plurality of storage bins being in communication with a retrieval conveyance system, receiving a plurality of selected storage bins from the plurality of storage bins at a processing area in communication with a programmable

motion device, grasping and moving selected objects out of a plurality of selected storage bins, and providing the selected objects to a plurality of destination bins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following description may be further understood with reference to the accompanying drawings in which:
- (2) FIG. 1 shows an illustrative diagrammatic view of a storage, retrieval and processing system in accordance with an embodiment of the present invention;
- (3) FIG. 2 shows an illustrative diagrammatic view of a portion of a retrieval conveyance system for use in the storage, retrieval and processing system of FIG. 1;
- (4) FIGS. 3A and 3B show illustrative diagrammatic views of an embodiment of a bin displacement system for use in a retrieval conveyance system of the invention;
- (5) FIG. 4 shows an illustrative diagrammatic exploded view of a box assembly for use as a storage bin or destination bin in accordance with various embodiments of the present invention;
- (6) FIG. 5 shows the assembled box tray assembly of FIG. 4;
- (7) FIG. 6A-6D show illustrative diagrammatic views of a further embodiment of a bin displacement system for use in further embodiments of the invention;
- (8) FIG. 7 shows an illustrative diagrammatic view of a processing station in a system of an embodiment of the present invention;
- (9) FIG. 8 shows an illustrative diagrammatic view from the perception system of FIG. 5, showing a view of objects within a bin of objects to be processed;
- (10) FIGS. 9A and 9B show an illustrative diagrammatic view of a grasp selection process in a storage, retrieval and processing system of an embodiment of the present invention;
- (11) FIGS. 10A and 10B show an illustrative diagrammatic view of a grasp planning process in a storage, retrieval and processing system of an embodiment of the present invention;
- (12) FIGS. 11A and 11B show an illustrative diagrammatic view of a grasp execution process in a storage, retrieval and processing system of an embodiment of the present invention;
- (13) FIG. 12 shows an illustrative diagrammatic view of a storage, retrieval and processing system in accordance with a further embodiment of the present invention involving a plurality of rows of processing stations; and
- (14) FIG. 13 shows an illustrative diagrammatic view of a storage, retrieval and processing system in accordance with a further embodiment of the present invention involving further retrieval conveyance systems and further processing stations.
- (15) The drawings are shown for illustrative purposes only.

DETAILED DESCRIPTION

- (16) In accordance with an embodiment, the invention provides a storage, retrieval and processing system for processing objects. The system includes a plurality of storage bins providing storage of a plurality of objects, a programmable motion device, and a plurality of destination bins. The plurality of storage bins are in communication with a retrieval conveyance system. The programmable motion device is in communication with the retrieval conveyance system for receiving the storage bins from the plurality of bins. The programmable motion device includes an end effector for grasping and moving a selected object out of a selected storage bin, and is adapted for movement along a first direction. The plurality of destination bins are provided in at least one linear arrangement along the first direction of movement of the programmable motion device.
- (17) With reference to FIG. 1, a system 10 of an embodiment of the present invention includes storage section 12 for storing a plurality of storage bins 14, a retrieval section 16, and a processing section 18 that includes a programmable motion device 20 and destination bins 22. Generally,

storage bins **14** are provided to the processing section **18** by a bin displacement mechanism and the retrieval section **16**. As further discussed below, the programmable motion device **20** (e.g., a robotic articulated arm) has a base that moves back and forth along a gantry **24** above the selected storage bin(s). The programmable motion device **20** is programmed to retrieve objects from the selected storage bin(s), and provide them to destination bins **22** in accordance with a manifest.

(18) The storage section **12** includes two rows of storage bins **14**, and the system knows what is in each bin, and where each bin is positioned along the two rows as further shown in FIG. **2**. When a particular storage bin **15** is selected, the system will actuate a bin removal mechanism **26** that travels along between the bins, and stops adjacent the selected bin as shown in FIG. **3A**. With reference to FIG. **3B**, the system will then move an urging member **28** of the mechanism **26** to push the selected bin onto a conveyor **30** of the retrieval section **16**, from which the selected storage bin will be directed to the processing section **18**. The removal mechanism **26** may actuate the urging member **28** by any of a variety of processes, including having the support beam **29** be threaded with the urging member **28** being threaded onto the beam **29** such that it moves when the support beam is rotated, or by other mechanical, pneumatic or electronic actuation.

(19) The conveyor **30** (as well as the other conveyors in the system) may be motion controlled so that both the speed and the direction of the conveyor (e.g., rollers or belt) may be controlled. In certain embodiments, the conveyors **30** and all of the conveyors of the retrieval section **16** may be gravity biased to cause any storage bin on any conveyor system to be delivered to the processing section **18**. In such a gravity fed system, when a bin is removed, the system will know that all bins uphill of the removed bin will move (e.g., roll) one bin lower on the conveyor. Further new bins may be manually or automatically added to the uphill end of the storage conveyor.

(20) The bins may be provided as boxes or containers or any other type of device that may receive and hold an item. In further embodiments, the bins may be provided in uniform trays (to provide consistency of spacing and processing) and may further include open covers that may maintain the bin in an open position, and may further provide consistency in processing through any of spacing, alignment, or labeling.

(21) For example, FIG. **4** shows an exploded view of a box tray assembly **130**. As shown, the box **132** (e.g., a standard shipping sized cardboard box) may include bottom **131** and side edges **133** that are received by a top surface **135** and inner sides **137** of a box tray **134**. The box tray **134** may include a recessed (protected) area in which a label or other identifying indicia **146** may be provided, as well as a wide and smooth contact surface **151** that may be engaged by an urging or removal mechanism as discussed below.

(22) As also shown in FIG. **4**, the box **132** may include top flaps **138** that, when opened as shown, are held open by inner surfaces **140** of the box cover **136**. The box cover **136** may also include a recessed (protected) area in which a label or other identifying indicia **145** may be provided. The box cover **136** also provides a defined rim opening **142**, as well as corner elements **144** that may assist in providing structural integrity of the assembly, and may assist in stacking un-used covers on one another. Un-used box trays may also be stacked on each other.

(23) The box **132** is thus maintained securely within the box tray **134**, and the box cover **136** provides that the flaps **138** remain down along the outside of the box permitting the interior of the box to be accessible through the opening **142** in the box cover **136**. FIG. **5** shows a width side view of the box tray assembly **130** with the box **132** securely seated within the box tray **134**, and the box cover holding open the flaps **138** of the box **132**. The box tray assemblies may be used as any or both of the storage bins and destination bins in various embodiments of the present invention.

(24) With reference to FIGS. **6A-6D**, a box kicker **184** in accordance with an embodiment of the present invention may be suspended by and travel along a track **186**, and may include a rotatable arm **188** and a roller wheel **190** at the end of the arm **188**. With reference to FIGS. **6B-6D**, when the roller wheel **190** contacts the kicker plate **151** (shown in FIG. **4**) of a box tray assembly **120**, the arm **188** continues to rotate, urging the box tray assembly **180** from a first conveyor **182** to a

second conveyor **180**. Again, the roller wheel **190** is designed to contact the kicker plate **151** of a box tray assembly **181** to push the box tray assembly **181** onto the conveyor **180**. Such a system may be used to provide that boxes that are empty or finished being unloaded may be removed (e.g., from conveyor **182**), or that boxes that are full or finished being loaded may be removed (e.g., from conveyor **182**). The conveyors **180**, **182** may also be coplanar, and the system may further include transition roller **183** to facilitate movement of the box tray assembly **181**.

(25) With reference to FIG. 7, the selected storage bin is received along a storage processing conveyor **32**. In various embodiments, the selected storage bin **14** may be processed fully and then discarded at an exit end **34** (shown in FIG. 1) of the conveyor **32**, or in certain embodiments, multiple selected storage bins may be processed in batches, with the empty boxes being discarded together. During processing, a perception unit **36** is attached to the device base, and looks down into the selected storage bin **14**. The end effector **40** of the programmable motion device **20** grasps an object in the bin **14**, and moves to deliver the object to a desired destination bin **22**. The programmable motion device **20** together with the grasped object may be moved along a gantry **35** to a desired destination bin **22**. Each of the objects in the selected storage bin **14** is provided to a destination bin **22** as required, and the bin **14** is then moved away from the processing area.

(26) In other embodiments, and in the event that the bin **14** is not emptied but processing of the bin is otherwise complete, the system may return the bin **14** to the storage section **12** along the retrieval section **16** in the reverse direction. In this case, the returned storage bin may be returned anywhere in a line of the bins (e.g., an end) as long as the system knows where the bin has been returned, and knows how each of the bins may have been moved when the selected storage bin was transferred to the conveyor **32**. The storage bins, for example, may be biased (e.g., by gravity) to stack against one of the ends of each row of bins. Once a destination bin is completed, the system may employ the programmable motion device **20** to push the completed bin onto an output conveyor **38**.

(27) FIG. 8 shows an image view **50** of the bin **14** from the perception unit **36**. The image view shows the bin **14** (e.g., on the conveyor), and the bin **14** contains objects **78**, **80**, **82**, **84** and **86**. In the present embodiment, the objects are homogenous, and are intended for distribution to different distribution packages. Superimposed on the objects **78**, **80**, **82**, **84**, **86** (for illustrative purposes) are anticipated grasp locations **79**, **81**, **83** and **85** of the objects. Note that while candidate grasp locations **79**, **83** and **85** appear to be good grasp locations, grasp location **81** does not because its associated object is at least partially underneath another object. The system may also not even try to yet identify a grasp location for the object **84** because the object **84** is too obscured by other objects. Candidate grasp locations may be indicated using a 3D model of the robot end effector placed in the location where the actual end effector would go to use as a grasp location as shown in FIG. 8. Grasp locations may be considered good, for example, if they are close to the center of mass of the object to provide greater stability during grasp and transport, and/or if they avoid places on an object such as caps, seams etc. where a good vacuum seal might not be available.

(28) If an object cannot be fully perceived by the detection system, the perception system considers the object to be two different objects, and may propose more than one candidate grasps of such two different objects. If the system executes a grasp at either of these bad grasp locations, it will either fail to acquire the object due to a bad grasp point where a vacuum seal will not occur, or will acquire the object at a grasp location that is very far from the center of mass of the object and thereby induce a great deal of instability during any attempted transport. Each of these results is undesirable.

(29) If a bad grasp location is experienced, the system may remember that location for the associated object. By identifying good and bad grasp locations, a correlation is established between features in the 2D/3D images and the idea of good or bad grasp locations. Using this data and these correlations as input to machine learning algorithms, the system may eventually learn, for each image presented to it, where to best grasp an object, and where to avoid grasping an object.

(30) As shown in FIGS. 9A and 9B, the perception system may also identify portions of an object

that are the most flat in the generation of good grasp location information. In particular, if an object includes a tubular end and a flat end such as object **87**, the system would identify the more flat end as shown at **88** in FIG. **9B**. Additionally, the system may select the area of an object where a UPC code appears, as such codes are often printed on a relatively flat portion of the object to facilitate scanning of the barcode.

(31) FIGS. **10A** and **10B** show that for each object **90, 92**, the grasp selection system may determine a direction that is normal to the selected flat portion of the object **90, 92**. As shown in FIGS. **11A** and **11B**, the robotic system will then direct the end effector **94** to approach each object **90, 92** from the direction that is normal to the surface in order to better facilitate the generation of a good grasp on each object. By approaching each object from a direction that is substantially normal to a surface of the object, the robotic system significantly improves the likelihood of obtaining a good grasp of the object, particularly when a vacuum end effector is employed.

(32) The invention provides therefore in certain embodiments that grasp optimization may be based on determination of surface normal, i.e., moving the end effector to be normal to the perceived surface of the object (as opposed to vertical picks), and that such grasp points may be chosen using fiducial features as grasp points, such as picking on a barcode, given that barcodes are almost always applied to a flat spot on the object.

(33) With reference again to FIG. **7**, destination bins **22** that are full or are otherwise finished being processed, may be moved to a respective output conveyor **38** for further processing, e.g., further packaging labeling or shipment. Such bins (e.g., box assemblies as discussed above), may be moved from a respective destination bin processing conveyor **33** to an output conveyor **38** by any of a variety of means, including having either human personnel move the bin, having the robot move the bin, or employing a box kicker **184** as discussed above with reference to FIGS. **6A-6D**. Such a box kicker **184** may, for example, be employed along each long side of the storage processing conveyor **32**, and be used to urge completed boxes from a conveyor **33** to an adjacent conveyor **33**.

(34) With reference to FIG. **12**, in accordance with a further embodiment, the system **100** may include multiple sets (e.g., **2**) of storage bins in rows with bin removal mechanisms at a storage section **102**, each of which is in communication with a retrieval section **104**, which in turn, is in communication with multiple (e.g., two) parallel processing sections **106**, each of which includes a programmable motion device **108** that runs along a gantry **110**.

(35) In accordance with a further embodiment and with reference to FIG. **13**, the system **200** may provide multiple sets (e.g., three) of storage bins in rows with bin removal mechanisms at a storage section **202**, each of which is in communication with a retrieval section **204**, which in turn, is in communication with multiple (e.g., three) parallel processing sections **206**, each of which includes a programmable motion device **208** that runs along a gantry **210**.

(36) Control of the overall system **10, 100** and **200** may be provided by a computer system **60** that is in communication with the bin removal mechanism, the conveyors, as well the programmable motion device **20**. The computer system **60** also contains the knowledge (continuously updated) of the location and identity of each of the storage bins, and contains the knowledge (also continuously updated) of the location and identity of each of the destination bins. The system therefore, directs the movement of the storage bins and the destination bins, and retrieves objects from the storage bins, and distributes the objects to the destination bins in accordance with an overall manifest that dictates which objects must be provided in which destination boxes for shipment, for example, to distribution or retail locations.

(37) Those skilled in the art will appreciate that numerous modification and variations may be made to the above disclosed embodiments without departing from the spirit and scope of the present invention.

Claims

1. An object processing system comprising: a storage section comprising a plurality of storage bins and a storage bin retrieval system, said plurality of storage bins providing storage of a plurality of objects, said plurality of storage bins provided in at least one input linear arrangement and being positionally biased to a first end of the at least one input linear arrangement such that when a selected storage bin is removed from the plurality of storage bins by the storage bin retrieval system remaining storage bins among the plurality of storage bins are brought closer together to fill a storage bin space left by the selected storage bin upon removal; a retrieval section for receiving the selected storage bin from the storage section, the retrieval section including at least one retrieval conveyor for moving the selected storage bin away from the storage section; a processing section for receiving the selected storage bin from the retrieval section, the processing section including a programmable motion device for grasping and moving a selected object out of the selected storage bin, the processing section further including a plurality of destination bins being provided in at least one output linear arrangement along an output direction adjacent the programmable motion device, wherein the programmable motion device is adapted to place the selected object from the selected storage bin into a selected destination bin among the plurality of destination bins; and a computer system that tracks an identity and a location of each storage bin among the plurality of storage bins, wherein the computer system updates the location of each storage bin within the remaining storage bins in response to the selected storage bin being removed from the plurality of storage bins.
2. The object processing system of claim 1, wherein the storage bin retrieval system includes an urging member that is adapted to travel along the at least one input linear arrangement.
3. The object processing system of claim 1, wherein the plurality of storage bins are positionally biased by gravity.
4. The object processing system of claim 1, wherein the programmable motion device is suspended from a gantry and is movable along the plurality of destination bins.
5. The object processing system of claim 1, wherein the programmable motion device is an articulated arm.
6. The object processing system of claim 5, wherein the articulated arm reciprocally moves between two rows of the plurality of destination bins, and the articulated arm is adapted to place the selected object into the selected destination bin.
7. The object processing system of claim 1, wherein the object processing system further includes a completed destination bin displacement system that urges a completed destination bin from among the plurality of destination bins onto the at least one output conveyor of the processing section.
8. The object processing system of claim 7, wherein the completed destination bin displacement system includes a box kicker that is adapted to travel along the output direction.
9. The object processing system of claim 1, wherein the plurality of destination bins are positionally biased to a first end of the at least one output linear arrangement such that when a completed destination bin is removed, remaining destination bins are brought closer together to fill an output bin space left by the completed destination bin.
10. The object processing system of claim 9, wherein the plurality of destination bins are positionally biased by gravity.
11. An object processing system comprising: a storage section comprising a plurality of storage bins and a storage bin retrieval system for retrieving a selected storage bin, the plurality of storage bins providing storage of a plurality of objects; a retrieval section for receiving the selected storage bin from the storage section, the retrieval section including at least one retrieval conveyor for moving the selected storage bin away from the storage section; a processing section for receiving the selected storage bin from the retrieval section, the processing section including a programmable

motion device for grasping and moving a selected object out of the selected storage bin, the processing section further including a plurality of destination bins being provided in at least one output linear arrangement along an output direction adjacent the programmable motion device, the programmable motion device being adapted to place the selected object from the selected storage bin into a selected destination bin among the plurality of destination bins, the processing section further including at least one output conveyor adjacent the at least one output linear arrangement of the plurality of destination bins, the at least one output conveyor for conveying a completed destination bin to a further processing location, the plurality of destination bins being positionally biased to a first end of the output linear arrangement such that when the completed destination bin is removed, remaining destination bins are brought closer together to fill an output bin space left by the completed destination bin; and a computer system that tracks an identity and a location of each destination bin among the plurality of destination bins, wherein the computer system updates the location of each destination bin within the remaining destination bins in response to the completed destination bin being removed from the plurality of destination bins.

12. The object processing system of claim 11, wherein the programmable motion device is suspended from a gantry and is movable along the plurality of destination bins.

13. The object processing system of claim 11, wherein the programmable motion device is an articulated arm.

14. The object processing system of claim 11, wherein the articulated arm reciprocally moves between two rows of the plurality of destination bins, and the articulated arm is adapted to place the selected object into the selected destination bin.

15. The object processing system of claim 11, wherein the plurality of destination bins are positionally biased by gravity.

16. The object processing system of claim 11, wherein the object processing system further includes a completed destination bin displacement system that urges a completed destination bin from among the plurality of destination bins onto the at least one output conveyor of the processing section.

17. The object processing system of claim 16, wherein the completed destination bin displacement system includes a box kicker that is adapted to travel along the output direction.

18. The object processing system of claim 11, wherein the storage section includes a storage bin removal system for removing the selected storage bin from the plurality of storage bins.

19. The object processing system of claim 18, wherein the plurality of storage bins are provided in at least one input linear arrangement and are positionally biased to a first end of the at least one input linear arrangement such that when a selected storage bin is removed from the plurality of storage bins by the storage bin retrieval system remaining storage bins among the plurality of storage bins are brought closer together to fill a storage bin space left by the selected storage bin upon removal.

20. The object processing system of claim 19, wherein the storage bin removal system includes an urging member that is adapted to move along the at least one input linear arrangement.

21. The object processing system of claim 20, wherein the storage bins are positionally biased by gravity.

22. A method of processing objects, said method comprising: storing a plurality of objects in a plurality of storage bins included in a storage section, said plurality of storage bins provided in at least one input linear arrangement adjacent to a conveyor of a retrieval section, displacing a selected storage bin from among the plurality of storage bins onto the conveyor of the retrieval system, said plurality of storage bins being positionally biased to a first end of the at least one input linear arrangement such that when the selected storage bin is removed, remaining storage bins are brought closer together to fill an input bin space left by the selected storage bin; updating a location of each storage bin within the remaining storage bins in a computer system responsive to the selected storage bin being removed from among the plurality of storage bins; receiving the selected

storage bin from the plurality of storage bins at a processing section that includes a programmable motion device and a plurality of destination bins in at least one output linear arrangement along an output direction; grasping and moving a selected object out of the selected storage bin and placing the selected object from the selected storage bin into a selected destination bin among the plurality of destination bins using an end effector of the programmable motion device; displacing a completed destination bin from among the plurality of destination bins onto at least one output conveyor of the processing section adjacent the at least one output linear arrangement of the plurality of destination bins, the at least one output conveyor conveying the completed destination bin to a further processing location using at least one output conveyor, the plurality of destination bins being positionally biased to a first end of the at least one output linear arrangement such that, when the completed destination bin is removed, remaining destination bins are brought closer together to fill an output bin space left by the completed destination bin; and updating a location of each destination bin within the remaining destination bins in the computer system responsive the completed destination bin being removed from among the plurality of destination bins.

23. The method as claimed in claim 22, wherein the plurality of storage bins is positionally biased by gravity.

24. The method as claimed in claim 22, wherein the programmable motion device is suspended from a gantry, and the method further comprises moving the programmable motion device to the selected destination bin along the output direction while suspended from the gantry.

25. The method as claimed in claim 22, wherein the programmable motion device is an articulated arm suspended from the gantry that reciprocally moves between two rows of the plurality of destination bins, and the articulated arm is adapted to place the selected object into the selected destination bin.

26. The method as claimed in claim 22, wherein the plurality of destination bins is positionally biased by gravity.

27. The method as claimed in claim 22, wherein displacing the selected storage bin from among the plurality of storage bins onto the conveyor of the retrieval section includes urging the selected storage bin onto the conveyor of the retrieval section with a storage bin displacement system.

28. The method as claimed in claim 27, wherein the storage bin displacement system includes an urging member that is adapted to travel along the at least one input linear arrangement.

29. The method as claimed in claim 22, wherein displacing the completed destination bin from among the plurality of destination bins onto the at least one output conveyor includes urging the completed destination bin from among the plurality of destination bins onto the at least one output conveyor with a completed destination bin displacement system.

30. The method as claimed in claim 29, wherein the completed destination bin displacement system includes a box kicker that is adapted to travel along the output direction.
